LinkedIn Job Market Analytics

Analysis of 6,200 data-role job postings (Data Analyst, Data Scientist, BI Analyst, Data Engineer, and related roles) across Indian companies, exploring hiring trends, in-demand skills, salary progression, and location patterns.

Tools Used

- **SQL Server (SSMS)** — data cleaning, deduplication, and 12 analytical queries
- **Python (Pandas, Matplotlib)** — data visualization, hosted on Kaggle Notebooks
- **Power BI Desktop** — interactive dashboard with KPI cards and charts

Dataset

- 6,293 raw job postings → cleaned to 6,200 unique records
- Columns: job title, company, industry, location, experience level, salary range (LPA), required skills, posting date, applicant count, job type, employment type
- Realistic messiness included: ~6% missing salary data, ~8% missing company size, inconsistent casing/whitespace in company names — cleaned using SQL and Power Query

Key Insights

1. **Top hiring companies:** Tata Consultancy (156 postings), HDFC Bank (153), and Capgemini (147) lead the market.
2. **Most in-demand skills:** SQL, Excel, Power BI, Python, and Statistics are the top 5 skills requested across postings — the exact toolkit covered in this project.
3. **Salary growth by experience:**

Level	Salary Range (LPA)	Share of Postings
Entry	₹4.5 – 8	37.9%
Mid	₹10.5 – 14.1	31.0%

Senior	₹20.6 – 24.1	23.4%
Lead	₹34 – 37.5	7.7%

Salary roughly doubles at each career stage.

4. **Location:** Hyderabad (811 postings) and Bengaluru (757) lead hiring volume; Chennai has solid local demand (383 postings).
5. **Hiring is steady year-round** — no major seasonal spikes or dips, ranging 300–380 postings/month.
6. **Work mode:** Onsite roles still dominate (44.3%), followed by Hybrid (36.9%) and Remote (18.9%).
7. **Entry-level roles are the largest single segment** (37.9% of all postings) — strong opportunity for early-career candidates.

Project Structure

- `linkedin_job_postings.csv` — raw dataset
- SQL scripts — cleaning and analysis queries
- Python notebook (Kaggle) — visualizations
- Power BI dashboard (.pbix) — interactive report

Dashboard Preview

Interactive Power BI dashboard featuring:

- Total job postings & average applicants (KPI cards)
- Top hiring companies (bar chart)
- Salary by experience level (clustered bar chart)
- Top in-demand skills (bar chart)